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Magnetic structure and evolution of the island of Santa Maria, Azores

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A palaeomagnetic study of the major volcanic units of Santa Maria reveals dominantly reverse characteristic magnetizations with a mean direction of $D = 179$, $I = -49$ and $\alpha_{95} = 2.1°$ (85 specimens from 20 sites). Sporadically occurring normal polarity remanences are interpreted as an overprinted component imposed by the island's terminal volcanic phase (the Feteiras Formation) now primarily exposed as strongly oxydized pyroclasts. The palaeomagnetic results are combined with available radiometric age estimates and local magnetic field information, and an integrated interpretation of all the data yields the principal conclusion that the Facho and Pico Alto volcanic sequences, covering the major part of the island, were laid down at around the Miocene/Pliocene boundary (~ 5 Ma ago), i.e. immediately succeeding the local Messinian sedimentary horizon. The Feteiras Formation is assigned to the Gauss normal polarity chron (3.4–2.5 Ma) in the Upper Pliocene. It is tentatively inferred that the Feteiras event represented the final tectonomagmatic link between the Gloria Fault and the East Azores Fracture Zone. The question of a realistic lower age assessment for the older (pre-Messinian) volcanic complex remains undecided.

1. Introduction

The ESE–WNW trending island chain of the Azores is located on a broad complex platform adjoining the Mid-Atlantic Ridge at around 39° N. Morgan (1972) and Schilling (1975) suggested that the archipelago represents magmatic activity associated with a hot spot or mantle plume. Figure 1 depicts the geographic location together with the main tectonic framework. The Azores–Gibraltar Fracture Zone (Menard, 1965) is a structural lineament linking the Mid-Atlantic Ridge and southwest Iberia. The fracture zone is thought to represent the Africa/Europe plate boundary, but when viewed in the context of the present plate tectonic theory it is enigmatic in several respects. Thus east of 20° W the tectonically active region covers a broad crustal belt, and the inconsistent orientation of nodal planes and slip vectors justifies the conclusion that no single major plate boundary is present there (Grimison and Chen, 1986). West of 20° W the suggested plate boundary is defined by a marked topographic low extending nearly due west for about 400 km, the Gloria Fault (Laugh-

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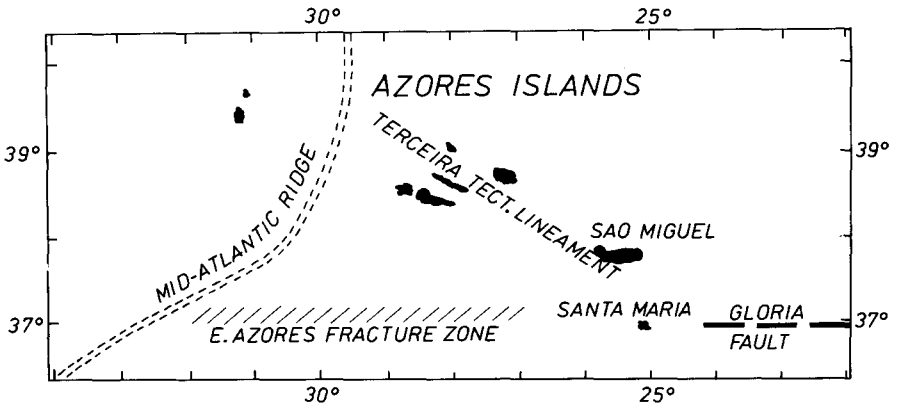


Fig. 1. Regional geographic and tectonic setting of Santa Maria.

ton and Whitmarsh, 1974). This structure has been interpreted as a transform fault (Laughton and Whitmarsh, 1974; Minster and Jordan, 1978) though it appears to have been devoid of significant seismic activity since at least about 1500 A.D. (Udias et al., 1976). It appears fair to conclude therefore that the transform nature claimed for the Gloria Fault has not been substantiated by relevant geophysical observations.

The Gloria Fault stops abruptly against the eastern submarine flank of Santa Maria, but it has been suggested (Krause and Watkins, 1970) that at some previous date the fault continued into the now aseismic East Azores Fracture Zone, originally bounding the European and African plates. At some time in the past the tectonomagmatic activity apparently shifted from the East Azores Fracture Zone to the presently seismically and volcanically active Terceira Rift (Krause and Watkins, 1970), a fault zone running in a WNW direction from the Gloria Fault (at 24° W) to the Mid-Atlantic Ridge. The major islands in the Azorean archipelago are situated along the Terceira Rift, and most of them are approximately elongated in a WNW–ESE direction like the archipelago itself.

The nodal position of Santa Maria, located close to the junction between the Gloria Fault, East Azores Fracture Zone, and the Terceira Rift, suggests that its volcanic history may hold further information with respect to the suggested regional change in tectonomagmatic pattern. The present paper basically reports on the palaeomagnetic build-up of the island, but available radiometric age determinations and magnetic field observations are also compiled, and an overall interpretation of all geophysical data is presented. The evidence sheds new light on the evolution of Santa Maria, but underlines also some previous suggestions that radiometric age determinations for establishing the history of volcanic islands may be extremely problematic. Recent geological studies of the island (Rodrigues et al., 1985; Madeira, 1986; Serralheiro et al., 1987) have been of utmost importance for the present investigation.

2. Geological evolution

The island of Santa Maria is the easternmost and apparently the oldest of the Azorean islands. Morphologically it can be divided into two parts: a western area defining a plateau-like topography with maximum elevations around 250 m above sea level, and an eastern area with more uneven topography and maximum heights of more than 450 m. Earlier geological studies (Zbyszewski and Ferreira, 1960, 1962a, 1962b) have recently been superseded by more detailed work (Rodrigues et al., 1985; Madeira, 1986; Serralheiro et al., 1987) which forms the basis for the present geophysical considerations. Figure 2 shows a simplified version of the geological map of Serralheiro et al. (1987).

The oldest rock unit exposed on the island, the Cabrestantes Formation, consists mostly of extremely altered submarine pyroclasts containing major amounts of large augite crystals. These rocks, which presently can be seen only in a minor outcrop on the north coast (not shown in Fig. 2), apparently formed during a transgressive period. This rise in relative sea-level stand was followed by a regression during which a substantial erosion of the Cabrestantes Formation took place, and the subaerial volcanism of the Porto Formation as well as that of the Anjos Formation came into existence. While the Porto Formation consists of subaerial cones and associated dykes, exposed in minor outcrops along the coast (not shown by Fig. 2), the overlying Anjos Formation, i.e. the youngest but by far the most dominant component of the old volcanic complex of Santa Maria, forms the foundation of the western part of the island. The Anjos lavas, which exhibit an average dip of around 15° in southwesterly directions, are cut by a subvertical dyke swarm with northeasterly strike directions.

The Anjos Formation is followed by an erosional and tectonic unconformity and a sedimentary series, the Fouril Complex, with conglomerates at the base and fossiliferous marine limestones and calcarenites at higher stratigraphic levels. The sediments near the base are intercalated with a subaerial lava flow and a submarine flow with pyroclastic material. The maximum thickness of the Touril Complex varies locally upwards to 100 m, and the fauna of the upper beds suggests an Upper Miocene age (Zbyszewski and Ferreira, 1962b). The dyke swarm that transects the Anjos lava sequence does not cut the Touril Complex.

The Touril sediments deposited during the transgression were followed by the onset of a new volcanic period during which the Facho and Pico

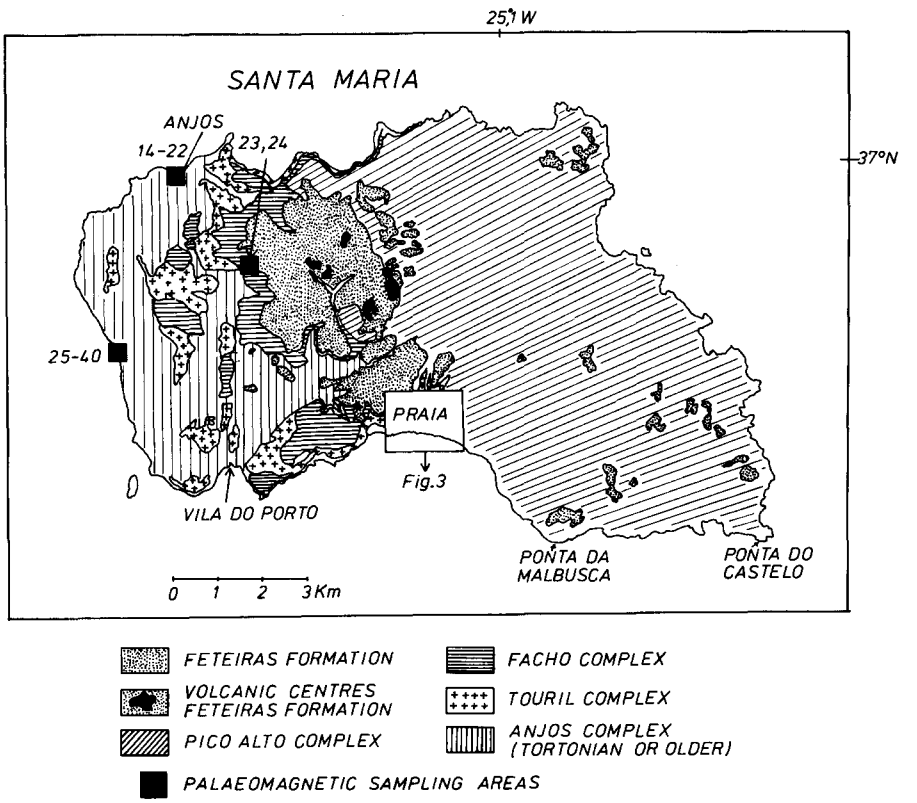


Fig. 2. Geological map of Santa Maria, simplified after Serralheiro et al. (1987).

Alto complexes, consisting mainly of horizontal to subhorizontal lava flows and pyroclastics, were laid down. After the initial transgression and submarine volcanism there was again a regression with associated erosion that formed terrigenous sediments (conglomerates) together with fossiliferous calcarenites. While the lower part of this volcanic succession, the Facho Complex, is concentrated in the western areas of the island, the younger part, the Pico Alto Complex, builds up the eastern half.

The final volcanic activity in Santa Maria is represented by the Feteiras Formation which essentially consists of subaerial pyroclasts (lapilli and ashes), but very small lava flows occur near the eruptive centres. The volcanic cones are strongly eroded and the pyroclasts are severely oxydized. At elevations of 140–200 m above sea level unfossiliferous beach deposits are found directly overlying the Feteiras Formations. In more recent times the island has been subjected to a steady regression and a number of thin Pleistocene–Quaternary beach deposits are found at various altitudes below 130 m. This may suggest that the youngest Santa Maria volcanism terminated in pre-Quaternary time (see also below).

3. Sampling and laboratory procedures

The rock material for the palaeomagnetic study was obtained by field drilling at 40 sites (including individual lavas, dykes etc.) and encompassing altogether 281 core samples. In the Praia area we sampled lavas from the Pico Alto, Facho and Touril complexes (sites 1–8), dykes from the Anjos Complex (sites 10–13), as well as a major collection (site 9) of conglomerate pebbles at the base of the Touril Complex (see Fig. 3 for further collection details). Two sites of pillow lavas of the Facho Complex (nos. 23 and 24) were sampled adjacent to the main exposure of pyroclasts of the Feteiras Formation. Two major stratigraphic profiles were collected in coastal sections of the Anjos Formation; at Anjos village, sites 14–22, and from the sea cliffs below the airport runway, sites 25–40

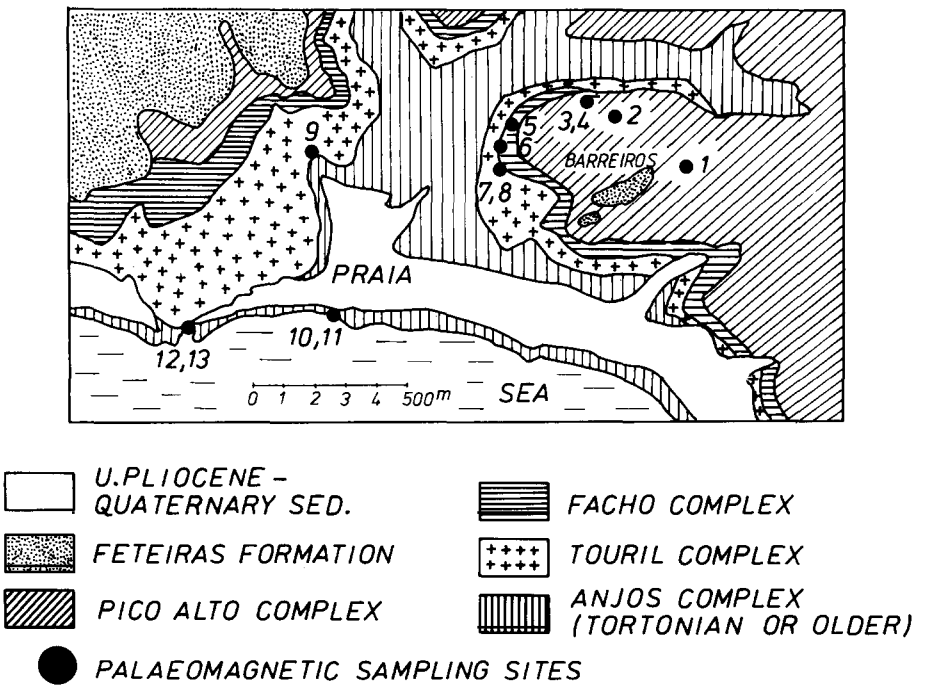


Fig. 3. Geological map of the Praia area (see Fig. 2) including location of sampling sites. The sampled material is as follows: sites 1–8, lavas; site 9, conglomerate at base of Messinian sediments (Touril Complex); sites 10–13, dykes.

(see Fig. 2). These two sections encounter lava flows, thin zones of baked brown to brick-red interbasaltic tuffs, as well as dissecting near-vertical dykes. The great majority of the samples have been oriented by both magnetic and sun compass orientation methods. The average declination based on the two types of measurement is 15.8° W as compared with the estimated regional declination of ~ 15° W. For individual samples the maximum declination variation was less than 5° from the mean value, so for samples that only have been magnetically oriented, the regional value of 15° W has been applied for declination corrections. The sampled material from seven sites, including an extensive collection of conglomerate pebbles from the Touril Complex (site 9), was lost in transit, but the remaining part of the collection (33 sites) has been subjected to detailed laboratory measurements and analysis.

For measuring remanent magnetism both a Digico spinner magnetometer and a Molspin magnetometer have been used. Alternating field (AF) demagnetization was carried out using two-axis tumblers, and for thermal demagnetization a Schonsted TM-1 was employed. All thermal work was done in Bergen while the bulk of the AF tests were carried out in Lisbon.

Previously published data giving vertical and horizontal components of the geomagnetic field for 15 observation points in Santa Maria (Quintino, 1960) have been used to construct a residual magnetic anomaly map. To obtain the residual value at individual points the average geomagnetic field for the island, based on the mean total field for all 15 stations, has been subtracted from the total field computed for each station.

4. Characteristics of remanent magnetization

Magnetization vectors have been defined from linear segments of orthogonal vector plots. Practically all of these magnetizations are those of highest stability (end-points), as consistent components of lower thermal or AF resistance have been successfully extracted from only three sites (nos. 1, 5 and 38). Of the 33 rock units (lavas, dykes and baked tuffs) analysed remanence vectors have been determined from altogether 25, the results of which form the basis for the palaeomagnetic discussion that follows. Specimens from the remaining eight sites exhibit unsystematic patterns during demagnetization and determination of their magnetization component(s) is unclear. The problematic rocks in this respect are all from the Anjos succession for which lavas and dykes are visibly more altered than material from the younger rock complexes.

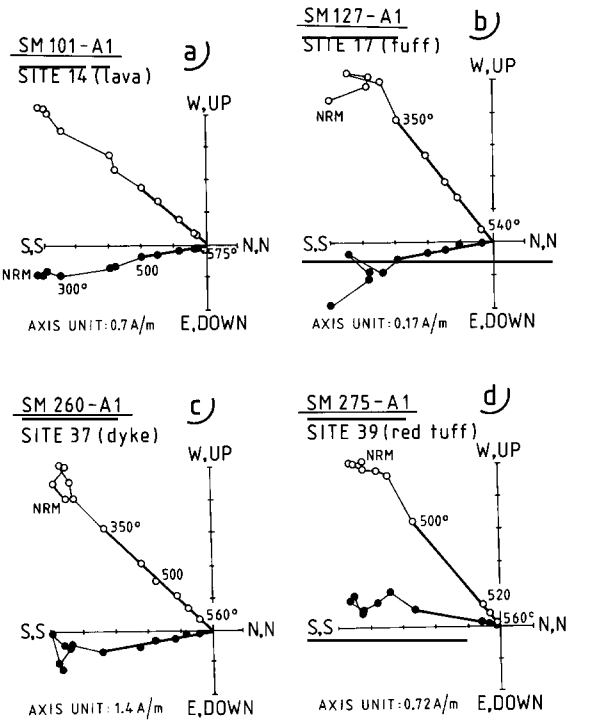


Fig. 4. Representative examples of stepwise thermal demagnetization results displayed in orthogonal vector plots. Open circles are end-point projections of the various demagnetized vectors, on a N–S vertical plane; solid circles, on the horizontal plane. The consistency of results regardless of rock type should be noted.

The fossil magnetization of our Santa Maria collection is dominantly of reverse polarity, and for the 25 sites discussed above linear trajectories in vector diagrams are very well displayed. Figure 4 presents examples from a lava, a dyke and two baked tuff horizons (all from the Anjos Formation). It is evident that all rock complexes studied here possess a reversely directed characteristic

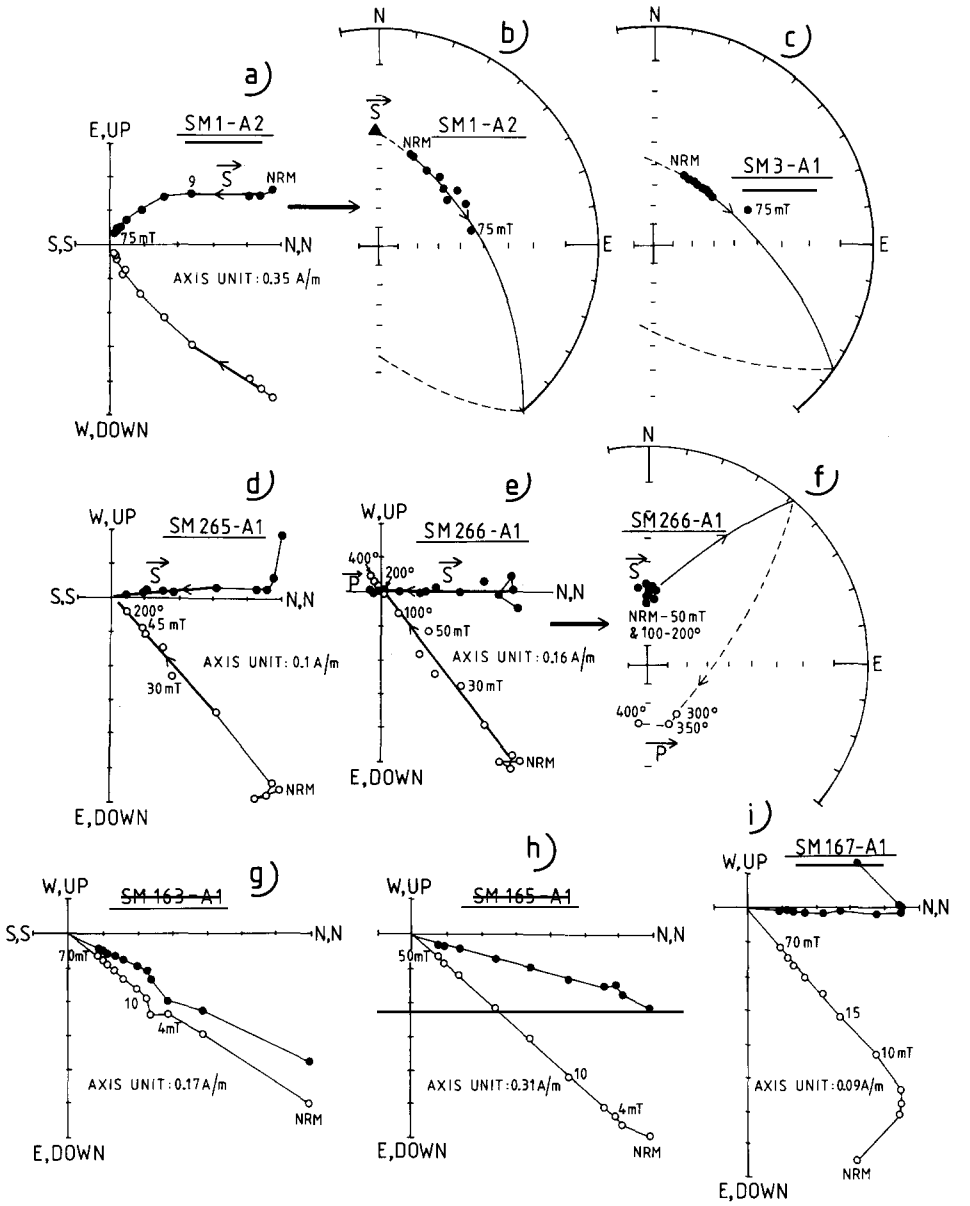


Fig. 5. Stepwise demagnetization of specimens from sites which display normal polarity magnetization. In the stereographic projections (b, c and f) solid circles denote lower hemisphere; open circles, upper hemisphere. Other symbol conventions are as for Fig. 4. In (b) the triangle represents the average vector subtracted direction (corresponding to the straightline segments in (a)) for the NRM-9 mT (milliTesla) range. At high alternating fields site 1 samples (a–c) have a strong component overlap of normal and reverse magnetizations, so that the total vector decays to zero before the most stable component (of reverse polarity) becomes isolated. For specimen SM 266-A1 (site 38) the two-polarity remanence are detected and resolved by first applying AF demagnetization to 50 mT and then subsequent thermal treatment to temperatures $\geqslant 300\,^{\circ}$C (e and f). The neighbouring specimen SM 265-A1(d) turned into an erratic stage on thermal treatment above $200\,^{\circ}$C and any underlying reverse component was not detected. For specimens that come from the two sites (23 and 24) situated adjacent to the main exposure of Feteiras pyroclasts (see g–i) no evidence for reverse magnetization has been found. We believe that these two Facho sites became magnetically completely reset during the Feteiras volcanic event. In cases where both normal and reverse magnetizations are seen to co-exist in individual specimens it is the reverse component that has the higher stability. Based on this and on the fact that the Santa Maria volcanics are predominantly reversely magnetized, we refer to the normal magnetization as secondary S and the reverse magnetization as primary P.

magnetization, with declinations nearly due south and intermediate inclinations upwards.

Normally directed magnetizations have been found in five sites. In three of these locations (nos. 1, 5 and 38) the normal component is apparently overprinted on the predominantly reverse magnetization (clearly demonstrated by SM 266-A1, Figs. 5e and f), while thermal and AF demagnetization did not uncover underlying reversed remanences in sites 23 and 24 (see Figs. 5g–i). The latter rocks are Facho Complex lavas located very close to the main occurrence of Feteiras pyroclasts. Also sites 1 and 5 have Feteiras pyroclasts nearby. The present palaeomagnetic collection was obtained before the recent detailed geological mapping, and we were unaware of the small lava flows associated with the Feteiras eruptive centres, which may be potential material for palaeomagnetic analysis. However, we believe that the Feteiras volcanism took place in a normal polarity epoch. The volcanic activity was of minor intensity so only a modest and irregular remagnetization of the host rocks took place. Sites 23 and 24 which are located very close to the Feteiras volcanic centres were apparently completely remagnetized while sites further afield were either not affected at all or they became sporadically overprinted due to the local supply of thermochemical agents. This hypothesis gains some support from the total magnetic field anomaly picture displayed in Fig. 6: the positive field anomaly has its maximum over central regions of the island which also cover the explosive centres of Feteiras volcanism. If this interpretation is correct the positive field anomaly must be primarily related to the subsurface part of Feteiras material as the shape of the magnetic anomaly does not follow the outcrop pattern (see also below).

In conclusion, the characteristic magnetization of the Anjos, Touril, Facho and Pico Alto Complexes is of reverse polarity, and the same field axis is apparently present in the various volcanic subdivisions. The normally directed magnetization, occurring sporadically in these rocks, was probably caused by remagnetization during the Feteiras magmatic event. The question then arises whether the Anjos rocks may have become magnetically reset during the extensive magmatism

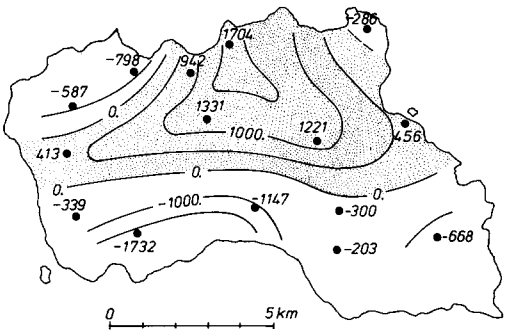


Fig. 6. First-order total field magnetic anomaly map based on 15 observation points of vertical and horizontal components of the field. Anomaly values are in nanoTeslas and are obtained by subtracting the average of the total field strength for all stations from the total field intensity at individual points. (See text for further discussion).

forming the Facho and Pico Alto complexes. Despite extensive and careful studies of the various rocks, notably the baked interbasaltic tuffs, we have not found any evidence for reverse magnetizations overprinting normally directed remanences. In some previous studies of the Canary and Cape Verde Islands (Storetvedt et al., 1979; Storetvedt and Løvlie, 1983) the presence of more than one axis of magnetization has been revealed from the older volcanic complexes. Such palaeomagnetic structures have not been found in the present study. Figure 7 shows the distribution at specimen level for assumed primary and sec-

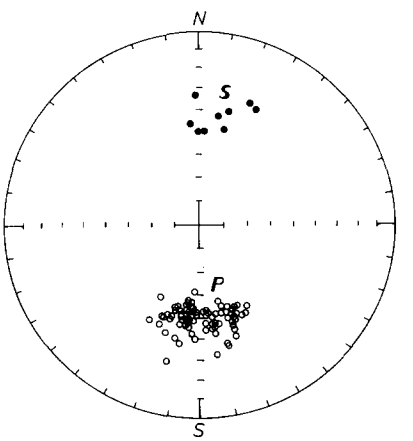


Fig. 7. Equal area plot of all individual specimen directions. Symbol conventions etc. are as for Fig. 5.

TABLE 1

Summary of in-situ characteristic magnetizations at site level for all single-polarity sites (parameters according to Fisher (1953))

Site	N	$\bar{D}$	$\bar{I}$	K	α_{95}
2	5	190	−54	184	5.7
3	4	185	−52	1520	2.4
4	6	187	−52	450	3.2
6	5	180	−47	274	4.6
7	3	177	−51	318	6.9
14	2	164	−48		
16	4	184	−46	31	16.7
17	2	189	−49		
18	4	176	−51	136	7.9
19	2	193	−46		
20	2	202	−54		
23	4	019	+37	117	8.5
24	2	009	+49		
25	1	162	−38		
28	3	176	−50	65	15.4
30	2	154	−47		
31	14	164	−47	113	3.8
32	6	165	−46	122	6.1
33	2	189	−43		
34	2	194	−33		
37	9	191	−48	66	6.3
39	7	187	−52	157	4.8

ondary magnetizations. Table 1 summarizes the characteristic magnetization directions at site level for all single-polarity sites (i.e. sites 1, 5 and 38 excluded), and Table 2 lists overall data for all directions included in Fig. 7, plus three normal directions from dual-polarity sites (i.e. SM1-A2, SM265-A1 and SM266-A1, Fig. 5).

5. History of volcanic activity

The richly fossiliferous sediments separating the older and younger volcanic sequences of Santa Maria, the Touril Complex, has remained an important age horizon in discussions of the volcanic history of the island. The limy sediment shows a microfauna of both planktonic and benthonic species (Mitchell-Thomé, 1979), and from various age assignments to these fossil assemblages (Cotter, 1892; Berthois, 1953; Ferreira, 1955; Krejci-Graf et al., 1958; Zbyszewski and Ferreira, 1962b) an Upper Miocene origin, Tortonian–Messinian (8–5 Ma) appears most adequate (Zbyszewski and Ferreira, 1962b).

Owing to lack of resolution of the K–Ar technique the prospects of establishing an adequate polarity time scale beyond ages of say 5 Ma is not very promising (McDougall, 1977). However, the present status of the age–polarity relationship suggests that the Gilbert reversed chron lasted from ~ 5.4 Ma to ~ 3.4 Ma (McDougall, 1977), and work in Iceland (McDougall et al., 1977) indicates that the reversed field polarity was again predominant beyond ~ 5.8 Ma. Despite the great uncertainties of this polarity pattern, involving not only the accuracy problem of the K–Ar technique itself, but also the perhaps much more serious problems of magnetic and radiometric resetting in major volcanic regions (such as Iceland), the dominating reverse polarity magnetization encountered in the Santa Maria rocks is compatible with a volcanic history that is partly slightly older (Anjos Complex) and partly slightly younger (Facho and Pico Alto complexes) than the Touril sedimentary succession of uppermost Miocene age.

Various attempts have been made to date the volcanic complexes of Santa Maria by using conventional K–Ar measurements (Abdel-Monem et al., 1975; Feraud et al., 1980; Feraud et al., 1984). Thus a total of 20 individual age estimates are presently available from samples scattered through the major subdivisions of the volcanic pile. Discussions by these authors have largely been focused

TABLE 2

Mean palaeomagnetic data for Santa Maria

Formation	N	D	I	K	α_{95}	Remarks
Anjos, Touril, Facho and Pico Alto	85	179	−49	54	2.1	Unit weight on spec.
Anjos, Touril, Facho and Pico Alto	20	180	−49	65	4.1	Unit weight on site
Overprinted magnetization	9	010	+42	53	7.1	Unit weight on spec.

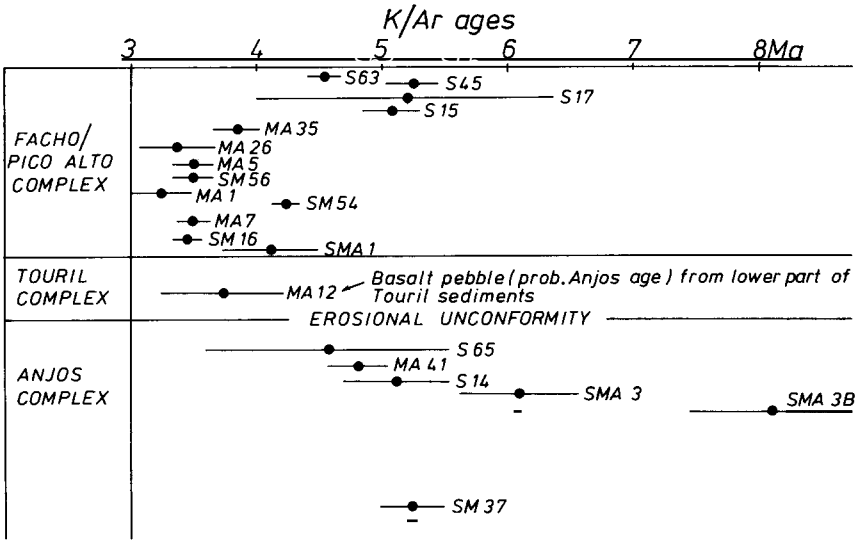


Fig. 8. Compilation of all published whole-rock K–Ar ages for Santa Maria including error limits. The inversion of apparent ages for the Facho/Pico Alto succession, and the significant spread of ages for Anjos rocks should be noted.

on the question of data reliability to certain age inconsistencies, but neither tests of age reproducibility within individual lavas or dykes nor a total assessment of the radiometric database have been provided. Based on the sample locations given in the individual studies we have compiled the existing age information versus approximate stratigraphic position in Fig. 8. For the Facho/Pico Alto complex we notice an inversion of apparent ages, the higher stratigraphic levels having ages ~ 5 Ma while the lower part of the sequence displays anomalously low apparent ages of 3.5 Ma. We interpret this pattern in terms of radiometric age resetting by the Feteiras magmatic event. With reference to the discussion of section 4 the Feteiras volcanism is likely to have occurred during a normal polarity interval of the geomagnetic field for which there are two obvious candidates: (1) the Gauss normal polarity chron, ~ 3.4–2.5 Ma, and (2) the Brunhes epoch, 0.7 Ma–present. In view of the fairly high elevations of the beach deposits immediately overlying the Feteiras volcanism (up to 200 m above sea level) and the evidence for systematic lowering of the relative sea level stand throughout subsequent times, it appears most appropriate to conclude that the Feteiras Formation dates back to the Gauss normal polarity epoch (Upper Pliocene). Based on this assumption it is inferred that the Facho succession and the lower part of the Pico Alto sequence suffered severe argon loss during the Feteiras magmatic event, providing a nearly complete resetting of the radiometric clock, while the upper part of the Pico Alto Complex apparently escaped this thermochemical overprinting. This implies that the ~ 5 Ma dates obtained from the Pico Alto rocks represent minimum ages for the Facho/Pico Alto volcanism. Therefore any age difference between these volcanics and the Touril Complex is likely to be small. This conclusion fits the geological evidence very well.

The only pronounced erosional and tectonic unconformity in Santa Maria is the one that has occurred between the formation of the Anjos volcanics and the Touril Complex. The Facho/Pico Alto volcanism represents a major magmatic event in the island, so it is very likely that thermochemical overprinting at that time altered or modified the original magnetic and radiometric characteristics of the basement volcanics. As can also be seen from Fig. 8, the available radiometric dates for Anjos rocks exhibit an excessive spread (from ~ 8 to 3.5 Ma) compared with the total age values, so the physical rock ages may be consider-

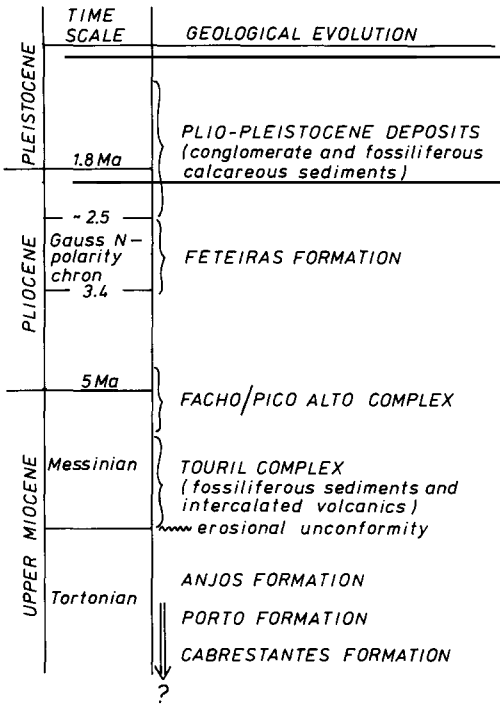


Fig. 9. Inferred evolutionary scheme for Santa Maria.

ably older than those estimated. Most likely we are here facing a similar problem to that encountered in Maio (Cape Verde Islands) where Upper Jurassic submarine volcanics (~ 150 Ma, based on fossil evidence) have been overprinted by Miocene thermochemical processes giving mixed radiometric ages in the 62–18 Ma age range, with a concentration of dates at the younger end (Mitchell et al., 1983; Storetvedt, 1987). Thus it is virtually impossible to establish the age of the Anjos Formation based on the whole-rock K–Ar results at hand. Neither can the palaeomagnetic results be of much assistance in fixing the age of these rocks which are relatively altered compared with the younger Facho/Pico Alto succession.

Figure 9 depicts the suggested evolutionary pattern. The new facets in this scheme are: (1) that the Facho/Pico Alto magnetism is placed in the uppermost Miocene, possibly following more or less directly upon the upper Touril volcanism, and (2) that the Feteiras event is inferred to have developed in the Upper Pliocene during the Gauss normal polarity chron. This implies that Santa Maria apparently has been devoid of active volcanism in say the last 2 million years, an anomaly with regard to all other islands in the archipelago. This conclusion can be further tested in the future through palaeomagnetic and K–Ar studies of the small lava flows occurring around the remnants of the Feteiras volcanic cones.

As can be seen from Fig. 6, the positive magnetic anomaly characterizing the island, interpreted here to be the surface expression of normally polarized rocks at depth (of Feteiras age), has a fairly marked E–W elongation. It is possible that we here see the link-up of the Gloria Fault (to the east) with the East Azores Fracture Zone (to the west). Furthermore, the northerly branching-off of the positive magnetic anomaly (Fig. 6) may reflect the switching of the tectonomagmatic activity away from the East Azores Fracture Zone and into the Terceira Rift in Upper Pliocene time.

6. Acknowledgements

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References

Abdel-Monem, A.A., Fernandez, L.A. and Boone, G.M., 1975. K–Ar ages from the eastern Azores group (Santa Maria, Sao Miguel and the Formigas Islands). Lithos, 4: 247–254.

Berthois, L., 1953. Contribution a l'etude lithologique de l'Archipel des Acores. Com. Serv. Geol. Portugal, 34: 198 pp.

Cotter, J.C.B., 1892. Noticia de alguns fosseis tertiarios da ilha de Santa Maria no archipelago dos Acores. Com. Trab. Geol. Portugal, 2: 255–287.

Feraud, G., Kaneoka, I. and Allegre, C.J., 1980. K–Ar ages and stress pattern in the Azores: geodynamic implications. Earth Planet. Sci. Lett., 46: 275–286.

Feraud, G., Schmincke, H.-U., Lietz, J., Gostaud, J., Pritchard, G. and Bleil, U., 1984. New K–Ar ages, chemical analysis and magnetic data of rocks from the islands of Santa Maria (Azores), Porto Santo and Madeira (Madeira archipelago) and Gran Canaria (Canary Islands). Arquipelago revista da Universidade dos Acores, Serie Ciencias da Natureza, 5: 213–240.

Fereira, O.V., 1955. Fauna Miocenica da ilha de Santa Maria (Acores). Com. Serv. Geol. Portugal, 36: 9–44.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London, Ser. A., 217: 295–305.

Grimison, N.L. and Chen, W.-P., 1986. The Azores–Gibraltar plate boundary: focal mechanism, depths of earthquakes, and their tectonic implications. J. Geophys. Res., 91: 1029–2027.

Krause, D.C. and Watkins, N.D., 1970. North Atlantic crustal genesis in the vicinity of the Azores. Geophys. J. R. Astron. Soc., 19: 261–283.

Krejci–Graf, K., Frechen, J., Wetzel, W. and Colom, G., 1958. Gesteine und Fossilien von den Azoren. Senckenbergiana Lethaea, 39: 303–351.

Laughton, A.S. and Whitmarsh, R.B., 1974. The Azores–Gibraltar plate boundary. In: L. Kristjansson (Editor), Geodynamics of Iceland and the North Atlantic Area. D. Reidel, Dordrecht, pp. 63–81.

Madeira, Y.E.O., 1986. Geologia estural e enquadramento geotectonico da ilha de Santa Maria (Acores). Relatorio Depart. Geol. Fac. Cien. Lisboa, 107 pp.

McDougall, I., 1977. The present status of the geomagnetic polarity time scale. Publ. No. 1288. Research School of Earth Sciences, Australia National University, Canberra, 34 pp.

McDougall, I., Saemundsson, K., Johannesson, H., Watkins, N.D. and Kristjansson, L., 1977. Extension of the geomagnetic polarity time scale to 6.5 Ma: K–Ar dating, geological and palaeomagnetic study of a 3500 m lava succession in western Iceland. Bull. Geol. Soc. Am., 88: 1–15.

Menard, H.W., 1965. Sea floor relief and mantle convection. Phys. Chem. Earth, 6: 315–364.

Minster, J.B. and Jordan, T.H., 1978. Present-day plate motions. J. Geophys. Res., 83: 5331–5354.

Mitchell, J.G., Le Bas, M.J., Zielonka, J. and Furnes, H., 1983. On dating the magmatism of Maio, Cape Verde Islands. Earth Planet. Sci. Lett., 64: 61–76.

Mitchell–Thomé, R.C., 1979. The pre-Quaternary stratigraphy of the Atlantic Islands. Geol. Rundschau, 68: 495–522.

Morgan, W.J., 1972. Deep mantle convection plumes and plate motions. Bull. Am. Assoc. Petrol. Geol., 56: 203–213.

Quintino, J.D.S. In: Servico Meteorologico Nacional 1960, 189 pp.

Rodrigues, B., Alves, C.A.M., Serralheiro, A. and Forjaz, V.H., 1985. Nota prévia sobre a petrologia e geoquimica na ilha de Santa Maria, Acores. Mem. Acad. Cien. Lisboa, 26: 71–91.

Schilling, J.-G., 1975. Azores mantle blob: rare-earth evidence. Earth Planet. Sci. Lett., 25: 103–115.

Serralheiro, A., Alves, C-A.M., Forjas, V.H. and Rodrigues, B., 1987. Carta Vulcanologia des Acoes. Ilha de Santa Maria (2 folhas), Esc. 1/15000. Centro de Vulcanologia—INIC, Lisboa.

Storetvedt, K.M., Mogstad Våge, H., Aase, S. and Løvlie, R., 1979. Palaeomagnetism and the early magmatic history of Fuerteventura (Canary Islands). J. Geophys. Res., 46: 319–334.

Storetvedt, K.M. and Løvlie, R., 1983. Magnetization properties of intrusive/extrusive rocks from East Maio (Rep. of Cape Verde) and their geological implications. Geophys. J. R. Astron. Soc., 73: 197–212.

Storetvedt, K.M., 1987. Evidence for ocean-continent crust boundary beneath the abyssal plain of the East Central Atlantic. Phys. Earth Planet. Inter., 48: 115–129.

Udias, A., Lopez Arrayo, A. and Mezcua, J., 1976. Seismotectonics of the Azores–Alboran region. Tectonophysics, 31: 259–289.

Zbyszewski, G. and Ferreira, O.V., 1960. Carta geologica de Portugal, Esc. 1/50000, Ilha de Santa Maria. Serv. Geol. Portugal.

Zbyszewski, G. and Ferreira, O.V., 1962a. Étude géologique de l'ile de Santa Maria (Acores). Com. Serv. Geol. Portugal, 46: 209–246.

Zbyszewski, G. and Ferreira, O.V., 1962b. La faune miocène de l'ile de Santa Maria (Acores). Com. Serv. Geol. Portugal, 46: 247–289.

Zbyszewski, G. and Ferreira, O.V. and Accumcao, E.T., 1961. Noticia explicativa da folha da ilha de Santa Maria (Acores). Serv. Geol. Portugal.